

Part A

Question	Question title	Question type
i	Instructions	Information or resources
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2	Q2	Upload Assignment
3	Q3	Upload Assignment
4	Q4	Upload Assignment

Part B

Question	Question title	Question type
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i Instructions

Complete your responses to this exam outside of Inspira. When you have done so, convert your responses to PDF files (one PDF per question). Upload your PDF answer files to the relevant question box. Upload one PDF per box.

This exam is in the so-called '*Handwritten/upload mode*'. This means that you have to produce the exam offline, either handwritten and scanned, or typed on a computer, or a mix of both.

Do not use your 30 minutes of technical time for continuing to write answers. The technical time is given to you to ensure you have enough time to convert your completed responses to PDFs and upload them. There is no grace period after the 30 minutes are up.

Download the exam paper [here](#).

Please leave this test window open throughout the exam.

Make sure to press 'Submit now' after (and only after) you have uploaded all your responses. Once you have pressed 'Submit now', you cannot edit your work any longer.

¹ Q1



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2 Q2

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4 Q4

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6 Q6

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7 **Q7**



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8 Q8

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9 Q9



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10 **Q10**



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SECOND PUBLIC EXAMINATION

Honour School of Philosophy, Politics, and Economics
Honour School of Economics and Management
Honour School of History and Economics

MICROECONOMICS

TRINITY TERM 2022

Wednesday 25 May
Opening time 14:30 UK Time

Mode of completion: Handwritten mode.

You have three hours writing time to complete the paper and up to 30mins technical time to upload your answer files. Please upload the answer to each question as a separate pdf file.

There are TEN questions in this paper.
Answer ALL Part A questions and TWO Part B questions.

Part A attracts 40% of the total marks. Each question in Part A has the same weight.
Each question in Part B attracts 30% of the total marks. For multipart questions, the weight assigned to each part is indicated in brackets.

The limit for essays is 1600 words, with technical materials (sensibly sized diagrams and/or equations, etc) contributing to the total at the rate of 350 words per A4 page.

You are permitted to use your own calculator.

Part A

1. Consider an exchange economy with two individuals, A and B, and two goods, x and y . Individual A has an endowment of $\omega_A = (5, 1)$ and individual B has an endowment of $\omega_B = (1, 5)$. The utility functions of the individuals are, respectively:

$$u_A(x_A, y_A) = x_A^2 y_A^3,$$

$$u_B(x_B, y_B) = \min\{2x_B, 3y_B\}.$$

- (a) Normalise the price of good x to 1, and denote by p the price of good y .
- (i) Verify whether the aggregate excess demand for good x at price p is equal to $\frac{10+2p}{5} + \frac{3+15p}{3+2p} - 6$.
 - (ii) How does the aggregate excess demand vary with p ? How is this related to the existence of a competitive equilibrium?
- (b) Consider now price $p = 1$.
- (i) What is the aggregate excess demand for good x at this price? Does this price correspond to an equilibrium? Is the corresponding equilibrium allocation Pareto efficient?
 - (ii) Illustrate with an Edgeworth box, including the endowments, the budget constraints and the relevant indifference curves of both individuals.

2. Two players, *Row* and *Col*, are playing the following game. Each player has the same available actions X and Y . The payoffs are symmetric, and represented in the matrix below, with the left payoff in each cell being that of *Row* and the right payoff in each cell being that of *Col*.

		<i>Col</i>	
		X	Y
<i>Row</i>	X	a, a	b, c
	Y	c, b	d, d

- (a) Consider first that the players move simultaneously.
- (i) If $a > c$ and $b > d$, what are the Nash equilibria? Is this a Prisoner's Dilemma?
 - (ii) If $a > c$ and $b < d$, what are the Nash equilibria, both in pure and mixed strategies?
- (b) Consider now that *Row* moves first, and *Col* knows the strategy that *Row* has chosen.
- (i) Explain the equilibrium concept that best applies to this type of situation.
 - (ii) If $b > d > c > a$, what is the equilibrium?

3. Lottery X pays prizes x_1 and x_2 with equal probability, while lottery Y pays prizes y_1 and y_2 with equal probability.
- (a) Assume that individuals like money. Explain whether the following statements are true or not:
- (i) If $y_1 < x_1 < x_2 < y_2$, any risk averse expected utility maximiser prefers X to Y .
 - (ii) If $y_1 < x_1 < x_2 < y_2$ and the expected value of X is above the expected value of Y , any risk averse expected utility maximiser prefers X to Y .
- (b) An individual, who again likes money, prefers lottery X to lottery Y . You offer this individual two modified lotteries: Compound lottery X' consists of a coin flip with the payment of lottery X when heads, and zero when tails; compound lottery Y' consists of a coin flip with the payment of zero when heads and the payment of lottery Y when tails. Explain whether the following statements are true or not:
- (i) If the individual is an expected utility maximiser, X' will be preferred over Y' .
 - (ii) If the individual maximises a utility function, X' will be preferred over Y' .

4. All households have the same wealth w , are risk-averse, and face a potential loss $K < w$. Half of the households are low-risk and have a probability $\frac{1}{4}$ of incurring the loss. Half of the households are high-risk, with a probability of $\frac{1}{2}$ of incurring the loss.

Insurance companies operate in a competitive market, are risk-neutral, and have no administrative costs.

- (a) Suppose that types are observable by insurers. Regulators 1 and 2 propose that insurance companies set different prices for different households, p_L and p_H per pound of cover. While Regulator 1 proposes that all households must fully insure, Regulator 2 proposes that they can partially insure (for no more than their total loss K). Explain why, in equilibrium, households may find both regulations equivalent. What are the equilibrium prices p_L, p_H ?
- (b) Regulators 3 and 4 propose that insurance companies must set the same price for all households, p per pound of cover.
- (i) Regulator 3 proposes that all households must fully insure. What is the equilibrium price p ?
- (ii) Regulator 4 proposes allowing households to partially insure (for no more than the total loss K). Denote by $q(p)$ the low-risk household's demand for insurance and assume that $q'(p) < 0$ holds for prices $\frac{1}{2} \geq p \geq \frac{1}{4}$. Verify whether there is a unique equilibrium price, which may be written as a weighted average of $\frac{1}{2}$ and $\frac{1}{4}$, at which insurance companies break even in expectation.

Part B

5. The senior tutor in an Oxford college wants to rank all the students in a group, in order to allocate a prize. In this ranking she wants to aggregate the opinions that different tutors have over these students. If you had to design a social choice rule, or some other mechanism, to help her with this task, which features should it have and why?
6. “Positive externalities have, by their nature, a positive impact on society and can therefore be disregarded by policy makers.” Discuss.
7. Suppose that two firms in a duopoly currently act non-cooperatively. They propose a merger. Outline how oligopoly theory can help the competition authority to determine the conditions under which the proposed merger should be allowed. How would your analysis of the merger of these two firms change if there were initially more than two firms in the industry (while it is still two firms who plan to merge)?
8. Suppose that you are a financial advisor, that all your clients are expected utility maximisers and that you want to give them individualised advice. What exact information, about their utility function over money and other relevant factors, is fundamental in order to understand their risk attitudes? Why is the Arrow-Pratt theorem relevant in this context? How would you design a survey using lotteries to obtain the desired information from client responses?
9. Consider a risk-neutral principal and an agent. Both the principal and the agent observe the realised profit level (which can be low or high), but only the agent knows the effort that they put in (which can be low or high). Evaluate the truth of the following statements, explaining your reasoning.
 - (a) There is a positive agency cost if and only if the principal chooses to induce high effort and the agent is risk-averse.
 - (b) With risk-averse agents, the more risk-averse the agent, the higher the agency cost.
 - (c) With risk-averse agents, the larger the difference between the high and low profit levels, the larger the difference between wages in the two states of the world when the principal wants to induce high effort.

10. College 1 and College 2 want to establish a joint library, with books available to members of both colleges. The preferences of both colleges are the same, and are represented by the utility function $u_i(B, x_i) = \frac{1}{3} \ln B + \frac{2}{3} \ln x_i$, where $B = b_1 + b_2$ is the total number of books in the joint library and x_i is the money that college i has for other uses. Each college has an initial endowment of £600 of money and no books. The price of books is $p_b = 1$ and the price of money is $p_x = 1$.
- (a) Suppose that each college decides, independently of the other, how many books to purchase and put in the joint library.
- Show that, if college 2 were to buy b_2 books, where $b_2 < 300$, college 1 would buy $b_1 = 200 - \frac{2}{3}b_2$ books.
 - How many books in total would there be in the library when each college chooses independently? [30%]
- (b) Consider now the optimal joint provision of books.
- Write down the condition that an optimal provision of books must satisfy.
 - What is the optimal number of books and why is it different from the number of books purchased when each college decides independently of the other? [30%]
- (c) Suppose now that the university wants to help colleges to implement the socially optimal number of books.
- Describe formally how the Lindahl pricing mechanism would work in this situation. (You may assume that in the social optimum each college pays for half the optimal number of books.)
 - Describe formally why the notion of strategy-proofness is relevant in this type of problem.
 - The university decides to let colleges act independently, and pays for a subsidy for books so the new price is below 1. What price will ensure that colleges buy the optimal number of books? [40%]